

Exercise: Restaurant Expenses

🔗 Objective

Design a simple but effective expenses recorder for a restaurant that will keep track of orders made to suppliers and will includes the following features:

- User Interface: Provide either a REST API or a CLI (your choice) to interact with the system.
- Data Ingestion: Allow users to submit supplier orders using either JSON or XML formats.
- Data Storage: Validate and store incoming orders in a local SQLite database (or another local DB of your choice).
- Querying: Allow users to query (via REST or CLI) recorded expenses by supplier name or by a specific time interval (from/to dates).

Example of JSON expenses to register

```
[
  {
    "supplier": "The Big Lobster",
    "issuedAt": "2026-07-04T11:30:00+02:00",
    "description": "Lobsters from Maine coasts",
    "qty": 25,
    "costPerUnit": 50,
    "category": "food"
  },
  {
    "supplier": "No more thirsty",
    "issuedAt": "2026-07-03T10:30:00+02:00",
    "description": "The Spumador ever ...",
    "qty": 500,
    "costPerUnit": 50,
    "category": "beverage"
  }
]
```

Example of XML expenses to register

```
<?xml version="1.0" encoding="UTF-8"?>
<expenses>
  <expense>
    <supplier>The Big Lobster</supplier>
    <issuedAt>2026-07-04T11:30:00+02:00</issuedAt>
    <description>Lobsters from Maine coasts</description>
    <qty>25</qty>
    <costPerUnit>50</costPerUnit>
    <category>food</category>
  </expense>
  <expense>
    <supplier>No more thirsty</supplier>
    <issuedAt>2026-07-03T10:30:00+02:00</issuedAt>
    <description>The Spumador ever ...</description>
    <qty>500</qty>
    <costPerUnit>50</costPerUnit>
    <category>beverage</category>
  </expense>
</expenses>
```

Evaluation

The goal of this assignment is not to build a flawless, production-ready system. Instead, we want to see you apply testing and architectural patterns that you can confidently explain and use to drive the discussion during your technical interview.